

Introduction to Electronics Engineering

MODULE 5 — PART 1: COMPLETE NOTES

Module 5 (08 hrs): Introduction to Electronic Communication Systems

For First-Year Students — written in simple, easy language
Course: Introduction to Electronics Engineering (IEE) · Course Code: EC 101

Module 5 — what it covers (as per the syllabus)

- Electromagnetic communication spectrum band and applications
- Elements of an electronic communication system
- Analog vs. digital communication
- Need for modulation
- Introduction to IoT
- Evolution of 1G to 5G and beyond

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1	The Electromagnetic Spectrum
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1. The Electromagnetic Spectrum

1.1 What is an Electromagnetic Wave?

An **electromagnetic (EM) wave** is a wave made up of an oscillating **electric field** and an oscillating **magnetic field** that travel together through space. The two fields are always at right angles to each other, and both are at right angles to the direction in which the wave travels.

The most important thing to remember about EM waves is that they do **not need any material medium** to travel — they can travel through empty space (a vacuum). This is exactly why radio and TV signals can reach us through the air, and why sunlight can reach the Earth through space. Light itself is just one type of electromagnetic wave. All EM waves travel at the same speed in a vacuum — the speed of light, $c = 3 \times 10^8 \text{ m/s}$.

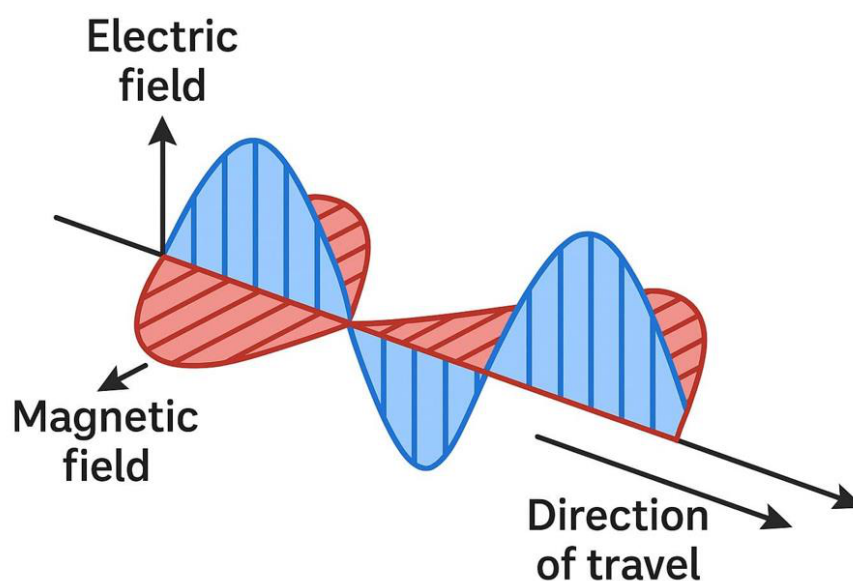


Figure 1: An electromagnetic wave — the electric and magnetic fields oscillate at right angles to each other.

1.2 What is the Electromagnetic Spectrum?

The **electromagnetic spectrum** is the complete family of all electromagnetic waves, arranged in order of their **frequency** and **wavelength**. Think of it as a long ruler of frequencies: at one end are waves with very low frequency (and very long wavelength), and at the other end are waves with very high frequency (and very short wavelength).

In order from **longest wavelength to shortest wavelength**, the bands are: **radio waves, microwaves, infrared, visible light, ultraviolet, X-rays and gamma rays**. Each band has its own characteristics and its own practical uses.

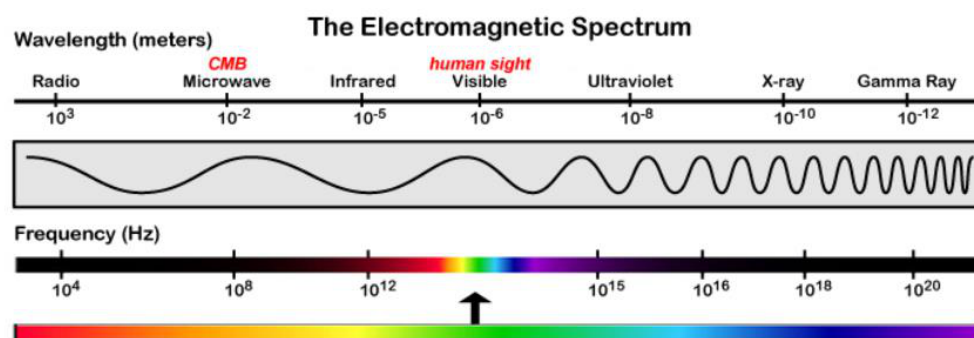


Figure 2: The electromagnetic spectrum and its applications.

1.3 Frequency and Wavelength

Wavelength (λ) is the distance between two consecutive peaks of the wave. **Frequency (f)** is the number of complete waves that pass a point every second (measured in hertz, Hz). The two are linked to the speed of light by a very simple and very important relation:

$$\lambda = c / f \quad (c = 3 \times 10^8 \text{ m/s})$$

This single relation explains the whole spectrum: if the frequency is **low**, the wavelength is **long**; if the frequency is **high**, the wavelength is **short**. So radio waves (low frequency) have very long wavelengths, while gamma rays (high frequency) have extremely short wavelengths.

Bands of the spectrum and their applications

Band	Typical Use
Radio waves	Broadcasting (AM/FM radio), television, mobile communication
Microwaves	Radar, satellite links, Wi-Fi, microwave ovens
Infrared	Remote controls, thermal imaging, optical fibre communication
Visible light	Human vision, optical fibre, photography
Ultraviolet	Sterilisation, medical treatment, detecting forged notes
X-rays	Medical imaging, security scanning
Gamma rays	Cancer treatment, sterilisation of medical equipment

Notice how the uses change with the band: radio waves are used for long-distance communication, microwaves for point-to-point links and radar, infrared for short-range control and fibre, and the very high-frequency X-rays and gamma rays for medical and industrial purposes.

2. Elements of an Electronic Communication System

2.1 What is a Communication System?

A **communication system** is a complete arrangement that carries information (a voice, a picture, a message, data) from one place — the **source** — to another place — the **destination**. It does this in five simple stages: the message is first converted into an electrical signal, then prepared for sending, then sent through a medium, then received, and finally converted back into a form a human can understand.

2.2 The Five Elements

- **Input transducer** — converts the message (sound, picture, data) into an electrical signal.
- **Transmitter** — converts that input signal into a transmitted signal that is suited to the transmission channel.
- **Transmission channel** — the electrical medium that bridges the distance from source to destination.
- **Receiver** — converts the received signal into a form appropriate for the output transducer.
- **Output transducer** — converts the output electrical signal back into the desired message form.

A simple way to remember the order is: **message** → **transducer** → **transmitter** → **channel** → **receiver** → **transducer** → **message**. The information goes out in one form and comes back in the same form at the far end.

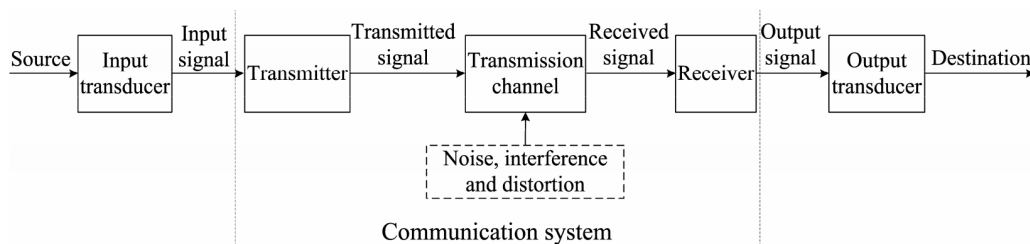


Figure 3: Block diagram of a basic communication system.

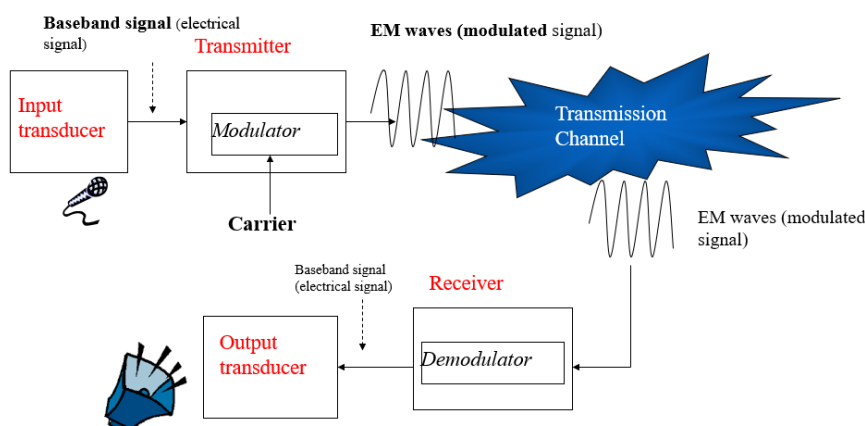


Figure 4: The transmitter, channel and receiver arrangement.

2.3 The Message

The communication system carries a **message**, and this message comes from the message source. Messages are of two broad types, and this difference is the whole basis of analog and digital communication:

- **Analog message** — one that is **continuous** in amplitude and over time. It can take any value in a range. Examples: AM/FM voice and sound, traditional TV video, and the output of a record player.
- **Digital message** — one that takes only **discrete values**, usually just 0 or 1. Examples: data stored on a disk, VCD/DVD, and the signals inside a cellular mobile phone.

Analog Message	Digital Message
Continuous in amplitude and over time	0 or 1, or a discrete value
AM, FM for voice/sound	Cellular mobile phone
Traditional TV for analog video	Data on your disk
Record player	VCD, DVD

Why does this matter? An analog signal changes smoothly and can be disturbed by noise easily, whereas a digital signal has only two levels, so it can still be recognised correctly even if noise slightly changes the voltage. This is the main reason modern systems have moved to digital.

2.4 The Transducer

A **transducer** is a device that converts energy from one form to another. In a communication system, transducers are the "bridge" between the real world and the electronic world: they turn physical quantities (like pressure, temperature, sound or light) into electrical signals, or electrical signals back into a physical effect.

- **Input transducers (sensors)** — convert a physical parameter into an electrical signal.
- **Output transducers (actuators)** — convert an electrical signal into a physical action.

Common examples: a **microphone** (sound → electricity), a **speaker** (electricity → sound), a **thermometer** (temperature → voltage), a **solar cell** (light → electricity), and an **ultrasonic probe**.

2.5 Transmitter and Receiver

Transmitter: a collection of electronic equipment designed to convert the message signal into a form suitable for transmission over a channel. Its typical operations are:

- 1 Encode the message data
- 2 Add a carrier signal (modulation)

In simple words, the transmitter first "packages" the data and then places it on a high-frequency carrier so that it can travel through the channel.

Receiver: a collection of electronic equipment designed to convert the message signal back to its original information. Its typical operations are:

- 1 Receive the signal
- 2 Remove the carrier signal (demodulation)
- 3 Decode the data
- 4 Remove noise

The receiver does exactly the opposite of the transmitter: it takes the signal off the carrier, decodes it, and cleans away the noise that the channel added, so that the original message is recovered.

2.6 The Communication Channel

The **communication channel** is the medium by which an electronic signal is transmitted from one place to another. Examples: conducting wires, coaxial cable, optical cable, and free space. There are two broad types:

Wireline Communication	Wireless Communication
Example: telegraph, telephone, fibre-optic communication	Example: mobile, radio and satellite communication
Two physical wires connect the transmitter and receiver	No physical wires connect the transmitter and receiver
Signal travels along the wire	Signal travels through free space as EM waves

3. Analog vs. Digital Communication

This is one of the most frequently asked comparisons. The table below lists every important difference, and the explanation that follows tells you the reason behind each one.

Feature	Analog Communication	Digital Communication
Signal	Continuous in amplitude and time	Discrete values (0 and 1)
Examples	AM, FM radio; traditional TV; record player	Cellular phones, data on disk, VCD/DVD
Noise immunity	Low — noise adds directly to the signal	High — noise can be largely removed
Bandwidth needed	Lower	Higher
Quality of reception	Reduced by noise	Improved (e.g. with PCM)
Security	Difficult to protect	Easy to encrypt

- An **analog** signal is a smooth, continuous curve. Because every value of the curve carries information, any noise that rides on the signal changes the message — the noise cannot be separated from the signal.

- A **digital** signal has only two levels, 0 and 1. As long as the receiver can still tell a 0 from a 1, the original data is recovered perfectly — so digital communication is far more immune to noise.
- Digital systems need more bandwidth, but they give better quality, easy encryption and the ability to carry data, voice and video together.

4. Modulation

4.1 Why do we need a carrier?

Imagine you have to send a small, light parcel a very long distance. You cannot throw it that far by hand — so you place it on a large, powerful truck and send the truck. In communication, the small parcel is the **message signal** (low frequency, weak) and the powerful truck is the **carrier signal** (high frequency, strong). We "load" the message onto the carrier and send the carrier through the channel. This loading process is called **modulation**.

4.2 What is Modulation?

Modulation is the process of varying one or more properties of a periodic waveform, called the **carrier signal**, with a modulating signal that typically contains the information to be transmitted. Two signals are involved:

- **$m(t)$ — the message signal**, also called the modulating signal or **baseband signal**. It is a **low-frequency** signal. The baseband is the band of frequencies of the source signal (e.g. audio 300 Hz–3.4 kHz; video 0–5 MHz).
- **$c(t)$ — the carrier signal**. It is a **high-frequency** signal, typically a sinusoid or a square pulse.

After modulation we get the **modulated signal $s(t)$** , which is what actually travels through the channel. At the receiver, the reverse process — **demodulation** — recovers the original message.

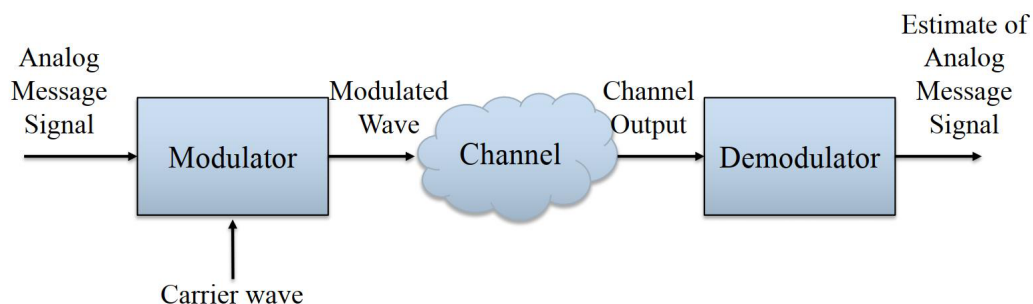


Figure 5: Modulation and demodulation — the message is placed on a carrier, sent through the channel, and recovered at the receiver.

4.3 Types of Continuous-Wave Modulation

A sinusoidal carrier has three properties we can vary: its **amplitude**, its **frequency**, and its **phase**. Varying each one in proportion to the baseband signal gives the three basic analog modulation techniques:

- **Amplitude Modulation (AM)**: the amplitude of the carrier is varied in proportion to the amplitude of the baseband signal, while the **frequency** of the carrier stays constant.
- **Frequency Modulation (FM)**: the frequency of the carrier is varied in proportion to the amplitude of the baseband signal, while the **amplitude** of the carrier stays constant.
- **Phase Modulation (PM)**: the phase of the carrier is varied in proportion to the baseband signal.

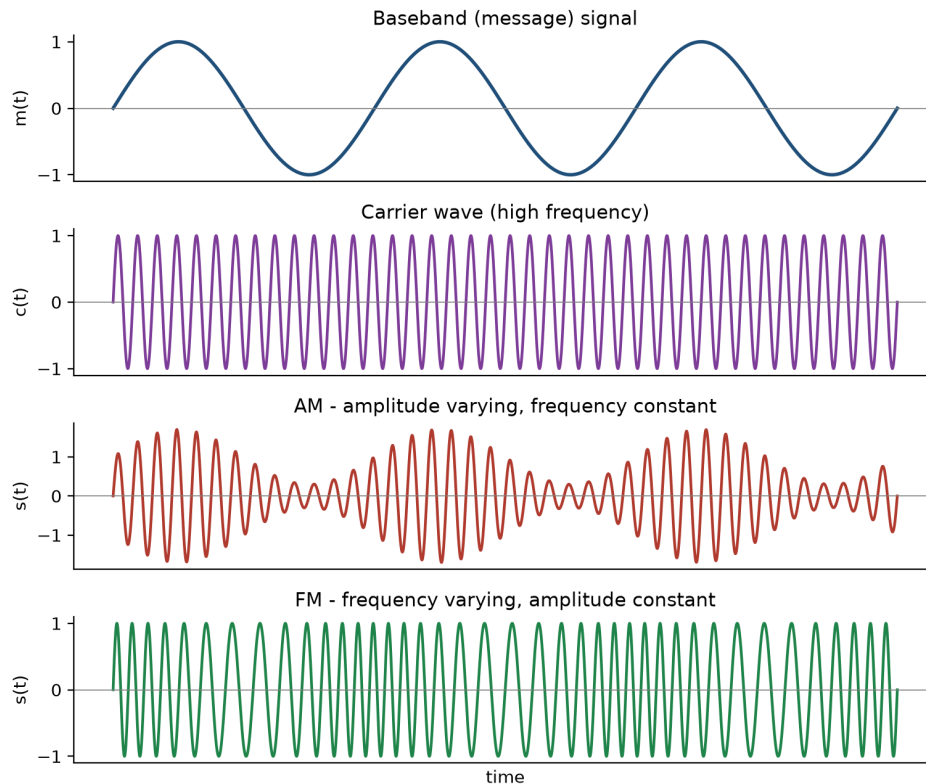


Figure 6: Baseband signal, carrier wave, AM wave (amplitude varying) and FM wave (frequency varying).

Look carefully at the AM wave: its peaks rise and fall in step with the message, but the distance between peaks stays the same. Now look at the FM wave: its peaks are equally tall, but they bunch together and spread apart in step with the message. That is the whole difference between the two.

4.4 Need for Modulation

Modulation is not optional — it is necessary for four main reasons:

- 1 Reduction in the height of the antenna
- 2 Increases the range of communication
- 3 Improves the quality of reception
- 4 Multiplexing is possible

(a) Reduction in the height of the antenna

For radio transmission, the antenna height must be at least a **quarter of the wavelength**, i.e. $\lambda/4$, where $\lambda = c/f$. Now suppose we try to send a baseband signal of $f = 10$ kHz directly:

$$\lambda = c/f = (3 \times 10^8) / (10 \times 10^3) = 30,000 \text{ m}$$

$$\lambda/4 = 30,000 / 4 = 7,500 \text{ m (about 7.5 km)}$$

An antenna of this height is practically impossible to install. But if we modulate the same signal onto a carrier of, say, 1 MHz, the wavelength becomes only 300 m and the antenna need be only 75 m — which is quite practical. **So modulation lets us use a short, practical antenna.**

$$\text{Minimum antenna height} = \frac{\lambda}{4} = \frac{c}{4f} = \frac{3 \times 10^8}{4 \times 10 \times 10^3} = 7500 \text{ meters i.e. 7.5 km}$$

Figure 7: Antenna height calculation.

(b) Increases the range of communication

The frequency of a baseband signal is low, and low-frequency signals cannot travel long distances — they get heavily attenuated (weakened) as they travel. The modulation process raises the frequency of the signal to be transmitted, and higher frequencies travel much further. Therefore modulation **increases the range of communication**.

(c) Improves the quality of reception

With frequency modulation (FM) and digital communication techniques such as pulse-code modulation (PCM), the effect of noise is reduced to a great extent. This **improves the quality of reception** — the recovered message sounds or looks much closer to the original.

(d) Multiplexing is possible

Multiplexing is a process in which two or more signals can be transmitted over the same communication channel simultaneously. This is possible only with modulation: each signal is placed on a **different carrier frequency**, so the signals do not interfere with one another and can be separated again at the receiver. Without modulation, all signals would overlap and become unusable.

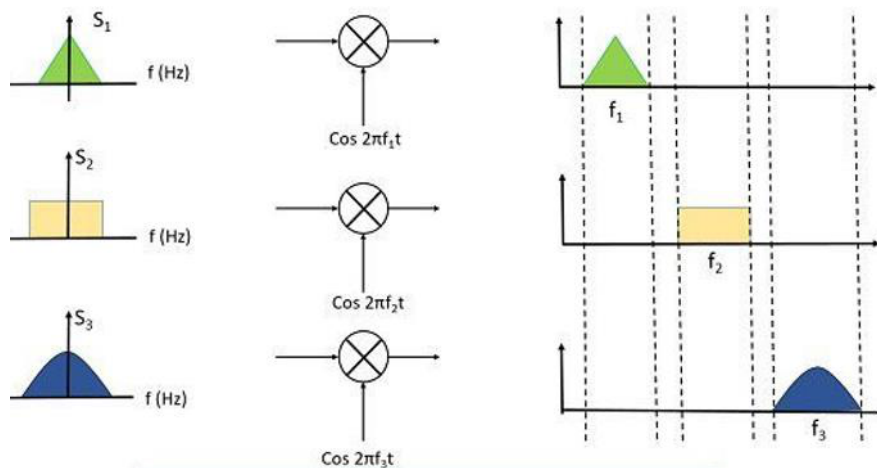


Figure 8: Frequency translation of several signals onto different carriers (the basis of multiplexing).

5. Introduction to the Internet of Things (IoT)

5.1 What is IoT?

The **Internet of Things (IoT)** is a system of interrelated, internet-connected objects that are able to collect and transfer data over a wireless network **without human intervention**. In simple words, IoT means giving everyday objects the ability to sense, connect and communicate on their own. The Internet of Things consists of a large network of interconnected devices.

Examples: smart fitness bands or watches, driverless cars or drones, smart homes that can be unlocked through smartphones, and smart cars.

5.2 Different Names of IoT

The same idea is known by many names in industry and research:

- Internet of Everything
- Smarter Planet
- Machine to Machine (M2M)
- Tsensors (Trillion Sensors)
- The Industrial Internet

- Industry 4.0
- Internet of Things (IoT)

5.3 How IoT Works

IoT works in four stages, one after another:

- 1 **Sensors/Devices** — sense the environment and collect data.
- 2 **Connectivity** — send that data to the cloud.
- 3 **Data Processing** — the cloud software analyses the data and makes it useful.
- 4 **User Interface** — deliver the useful information to the user.

The sensors are placed in devices and objects, and they feed data to various IoT platforms. The IoT platforms gather the pinpointed information and detect patterns. Thus IoT helps organisations and institutions reduce cost through improved process efficiency, asset utilisation and productivity.

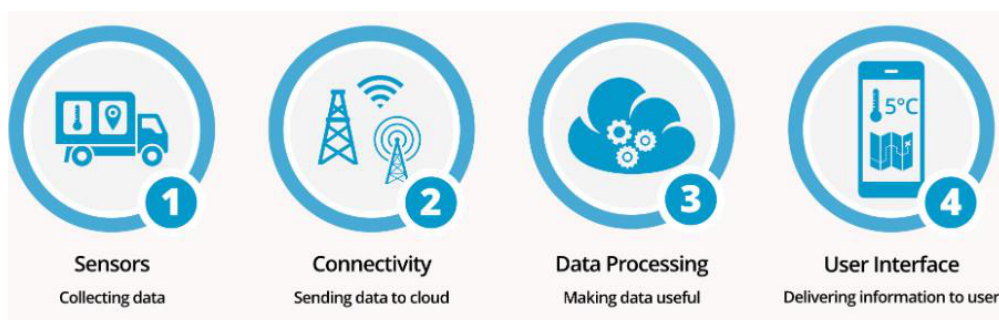


Figure 9: The four stages of IoT — sensors, connectivity, data processing and user interface.

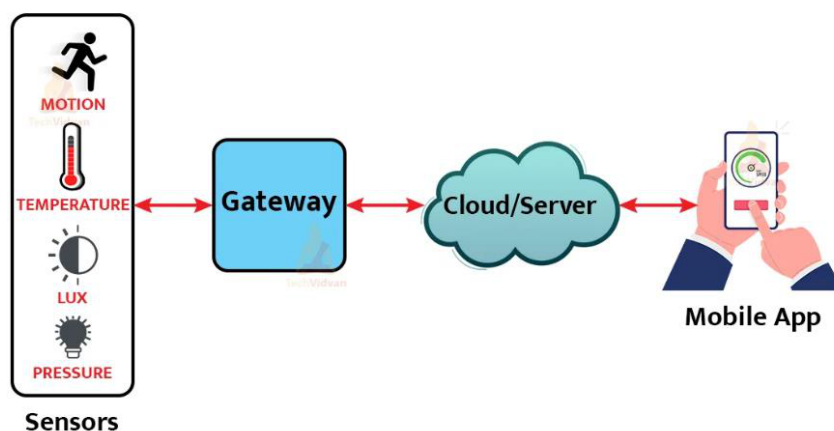


Figure 10: IoT data flow: sensors → gateway → cloud/server → mobile app.

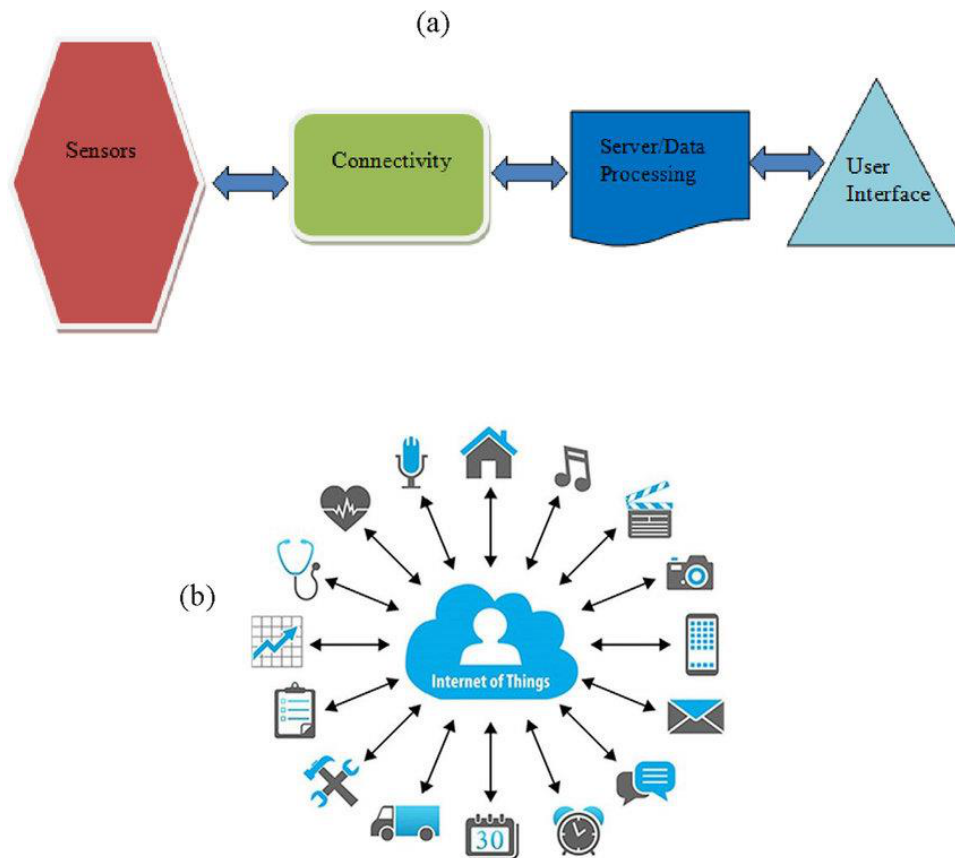


Figure 11: Structure of the Internet of Things and the different types of sensors.

5.4 Components of IoT (in detail)

- 1. Sensors:** these devices connect with the external physical environment. They collect data from outside changes and store this information. A sensor senses the changes in its surroundings and notes them down — this property makes sensors extremely useful in IoT applications. For example, there are sensors in your phone such as GPS, which track your location and guide you to your destination.
- 2. Connectivity:** all the collected data is sent to a cloud infrastructure. The sensors must be connected to the cloud using various mediums of communication, such as mobile or satellite networks, Bluetooth, Wi-Fi and WAN.
- 3. Data Processing:** once the data is collected and reaches the cloud, the software performs processing on the gathered data. This process can be as simple as checking the temperature or reading devices like ACs or heaters, or as complex as identifying objects using computer vision on video.
- 4. User Interface:** the information must be made available to the end-user in some way — by triggering alarms on their phones, or by sending notifications through email or text message. Sometimes the user needs an interface that actively checks their IoT system, for example a home camera whose recordings and feeds are accessed through a web server.

5.5 Advantages of IoT

- Eliminates or reduces the chances of human error.
- Decreases the wastage of resources — time, material and human effort.
- Boosts confidence and productivity.
- Increases security and accuracy.
- Enhances the quality of products and services.

5.6 Disadvantages of IoT

- If the interconnected devices are hacked, the entire system security may be jeopardised.
- The initial investment on hardware is heavy.
- Bringing in IoT will absorb many jobs that were done manually.
- In case of a fault in one device, the entire system may be disrupted.
- Too much dependence on IoT will decrease confidence in human effort.

5.7 IoT Applications

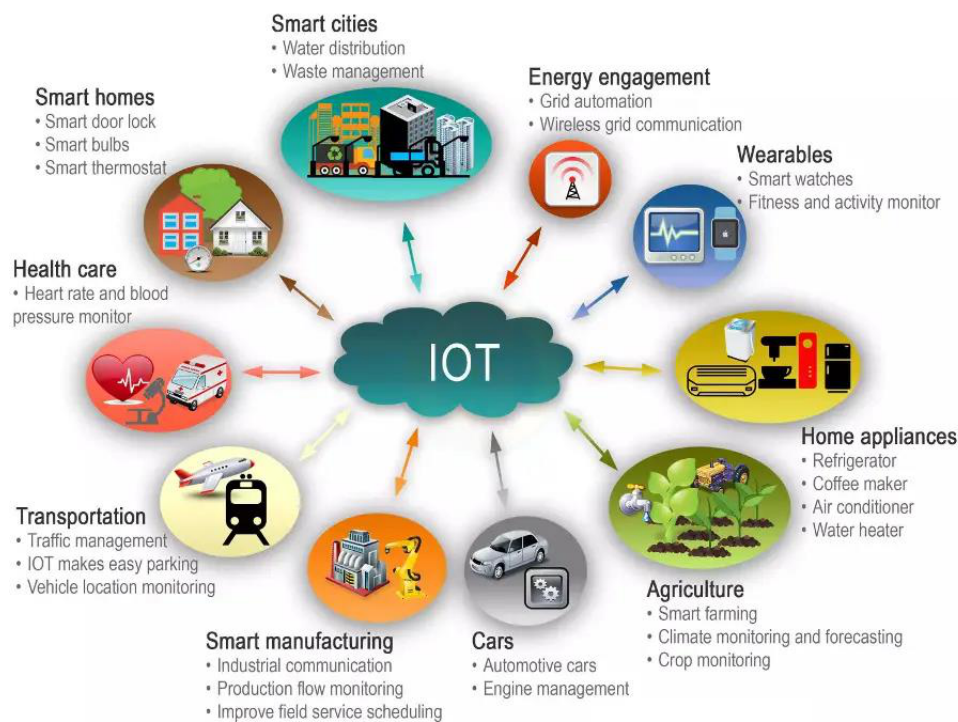


Figure 12: Common application areas of the Internet of Things.

Typical application areas include smart homes, smart wearables, healthcare monitoring, driverless vehicles and drones, industrial automation, agriculture and smart cities.

6. Evolution of Mobile Communication — 1G to 5G

G stands for **Generation** — a generation of wireless phone technology. Every new generation brought a faster data rate, better quality and completely new services. Following the journey from 1G to 5G is a very common question, so read each generation carefully.

6.1 First Generation (1G)

- Period: 1980 to 1990.
- Frequency: 150 MHz / 900 MHz; bandwidth: analog (30 kHz).
- Characteristic: the first wireless communication.
- Technology: analog cellular.
- Capacity (data rate): 2 kbps.
- Drawbacks: bad voice quality, poor battery, big cellphones.
- Better than nothing — at least it was wireless and mobile.



Figure 13: Early mobile communication devices.

6.2 Second Generation (2G)

- Period: 1991 to 2000.
- Frequency: 1.8 GHz (900 MHz); bandwidth: 900 MHz (25 MHz).
- Characteristic: digital; technology: digital cellular, GSM.
- Capacity (data rate): 64 kbps.
- Allows text-message service; the signal must be strong, otherwise a weak digital signal results.
- **2.5G** — 2G cellular technology with GPRS: e-mails, web browsing and camera phones.



Figure 14: A 2G mobile phone.



Figure 15: 2G-era handsets.

6.3 Third Generation (3G)

- Period: 2000 to 2010.
- Frequency: 1.6 – 2.0 GHz; bandwidth: 100 MHz.
- Characteristic: digital broadband, increased speed.
- Technology: CDMA, UMTS, EDGE.
- Capacity (data rate): 144 kbps – 2 Mbps.
- Features: called smartphones; video calls; fast communication; mobile TV.
- Drawback: 3G phones were rather expensive.



Figure 16: A 3G smartphone.

6.4 Fourth Generation (4G)

- Period: 2010 to today.
- Frequency: 2 – 8 GHz; bandwidth: 100 MHz.
- Characteristic: high speed, all IP.
- Technology: LTE, Wi-Fi.
- Capacity (data rate): 100 Mbps – 1 Gbps.
- MAGIC features: Mobile multimedia; Anytime, anywhere; Global mobile support; Integrated wireless solutions; Customized personal service.
- Good QoS and high security, but bigger battery usage.

6.5 Fifth Generation (5G)

- Capacity (data rate): 1 Gbps and beyond.
- High speed and capacity; faster data transmission than 4G.
- Supports interactive multimedia and voice streaming.
- More efficient than 4G.
- Frequency band: 450 MHz–6 GHz and 24.25–52.6 GHz.



Figure 17: 5G mobile communication.

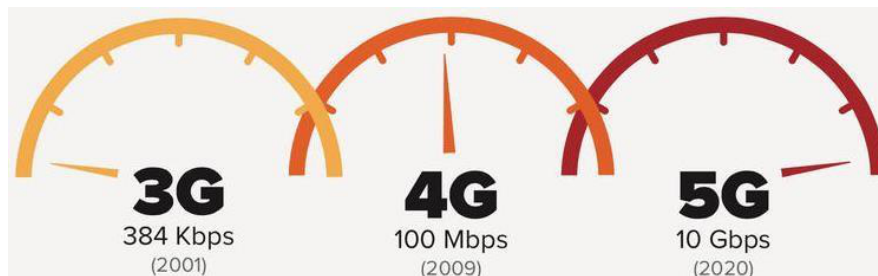


Figure 18: Data-rate growth across the mobile generations.

6.6 Comparison of 1G to 5G

Feature	1G	2G	3G	4G	5G
Period	1980–1990	1990–2000	2000–2010	2010–2020	2020–2030
Frequency band	850/1900 MHz	900/1800/1900 MHz	900/1800/1900/2100 MHz	1.8–2.6 GHz	450 MHz–6 GHz; 24.25–52.6 GHz
Bandwidth	Analog (30 kHz)	200 kHz (digital)	5 MHz	1.4–20 MHz	1000× bandwidth per unit area
Data rate	2 kbps	64 kbps	144 kbps–2 Mbps	2–100 Mbps	More than 1 Gbps
Multiplexing	FDMA	TDMA/CDMA	WCDMA	OFDM	Scalable OFDM, NOMA, FBMC
Switching type	Circuit	Circuit	Circuit / Packet	Packet	Packet
Application	Voice	Voice, SMS, low data	Voice, data, video	High-quality video	IoT-based communication
Technology	Analog cellular	Digital cellular (GSM)	CDMA, UMTS, EDGE	LTE, WiMax, Wi-Fi	Virtual Reality applications

Evolution of mobile phone communications

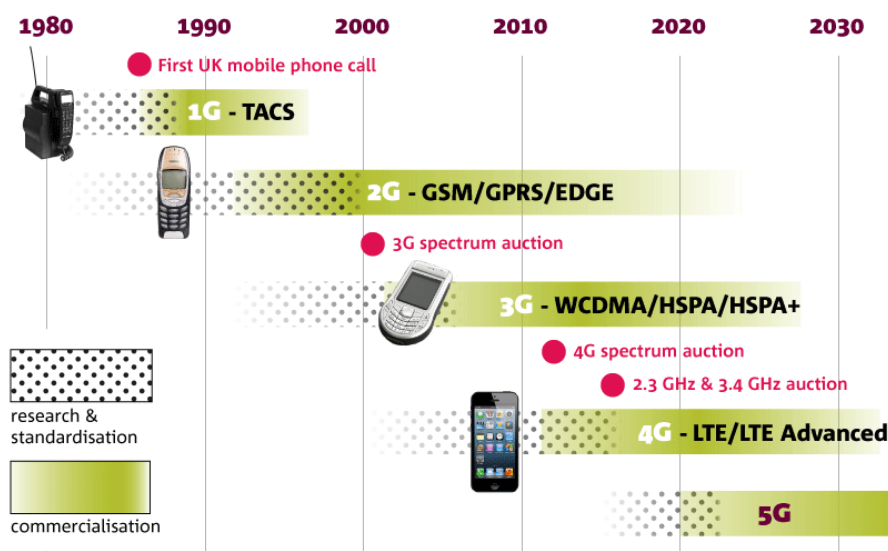


Figure 19: Timeline of the evolution of mobile phone communication.

7. Quick Reference — Key Points and Definitions

7.1 Key Definitions

Term	Definition
Electromagnetic spectrum	The complete range of electromagnetic waves arranged by frequency and wavelength.
Transducer	A device that converts energy from one form to another (e.g. microphone, speaker).
Transmitter	Equipment that converts the message signal into a form suitable for transmission (encodes data, adds the carrier).
Receiver	Equipment that recovers the original message (receives, demodulates, decodes, removes noise).
Communication channel	The medium by which the signal travels from transmitter to receiver (wire or free space).
Modulation	The process of varying a property (amplitude/frequency/phase) of a high-frequency carrier in accordance with the message signal.
Baseband signal	The band of frequencies of the source (message) signal — a low-frequency signal.
Carrier signal	A high-frequency sinusoidal or square-pulse signal used to carry the message.
AM / FM	Amplitude Modulation (amplitude varies) / Frequency Modulation (frequency varies).
Multiplexing	Transmitting two or more signals over the same channel simultaneously — possible only with modulation.
IoT	A system of interrelated, internet-connected objects that collect and transfer data over a wireless network without human intervention.
1G to 5G	The five generations of wireless phone technology, from analog voice (1G) to IoT-based high-speed communication (5G).

7.2 Key Formulae

Quantity	Formula	Notes
Wavelength	$\lambda = c / f$	$c = \text{velocity of light} = 3 \times 10^8 \text{ m/s}$
Minimum antenna height	$\lambda / 4 = c / (4f)$	Antenna must be a quarter-wavelength long